The Four Locks: Pricing Instrumentation, Casting the Seats, and Making Adaptation Measurable

Geist Atlas · Module 17 · What the Tutor Needs to Know

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[**SCOPE-BOUND**] Supported within a narrow scope · **Family:** What the Tutor Needs to Know

Claim: A frozen-replay pricing found exactly one advisory block pays on a plan-blind reading (one sentence of hidden state), and its free-running outcome trial died at a saturated gate. The follow-on arc located four structural locks — a stateless learner, guard enforcement that authors the tutor's hardest turns, a tutor seat whose refusal capacity is family-fixed, and turn-local judging that cannot see a cross-turn relation — and opened each with a priced mechanism. A pre-registered claim gate then failed for a temperament mechanism and passed at its weakest reading for per-turn move injection; naming the learner's state to a judge near-doubled its agreement with authored gold in two judge families.

One clause of information placement moved bare-tutor closure more than any tutoring behaviour observed (5/5 to 4/5 on a minimal-pair world). Planted, typed learner states with human-adjudicated repairs made repair delivery a measured quantity; a deterministic pressure trigger graduated from 6/20 to 17/20 against that gold in one tuning pass. The move-card mechanism delivered 63% right-repairs on covered moments against the baseline's 42%, with closures and leak discipline intact — k=3, one world, one persona, simulated learner, no human-learning claim. The disclosure result is the instrument finding: turn-local rubrics under-measure adaptation by construction, and one added sentence repairs half

*This sentence is the only one actually authored by Liam Magee in this paper.

the deficit.

Derived view of `paper-full-2.0.md` v3.0.299 — §6.23, §6.24. The paper is canonical; edit it there, not here.

6.23 Pricing the Speaking Tutor's Instrumentation on a Ruler the Plan Cannot Win: One Sentence of Hidden State Carries the Independent-Channel Effect (Post-Hoc, Development-Tier)

The tutor stub's speaking turn receives its private planner context as labelled advisory blocks in the final user message — a committed/due evidence window, a learner classifier, the redacted learner proof-DAG, the human discourse scaffold, and the per-turn first-draft performance contract of §6.18. Nothing had priced any single block, because varying instrumentation previously meant rerunning the dialogue and losing the comparison. A frozen-replay harness closes that gap: recorded dialogues on three authored worlds (Tallow Street, Nocturne, Greyfen; 14 recorded turns) are replayed past versions of the tutor that differ *only* in which blocks reach the speaker — public prefix, learner utterances, world, and evidence state frozen; every version graded by the recording's own pinned deterministic guard set, so a bare version is still judged on leakage, clue release, composition, and actorial realization (`services/tutorStubAbHarness.js`, `config/tutor-stub-ab.yaml`; runs and verdicts under `exports/tutor-stub-ab/`).

On the bench's own rules, exactly one block pays. Pooled over every recorded run, the bare tutor breaks 5.04 rules per turn; the contract alone breaks 1.67; every other single block sits at 4.4–5.3, indistinguishable from bare against a per-run noise band of ±3. But most of those rules ask whether the tutor did what its host plan told it to do, so the version holding the plan wins them close to by construction. A rule-keying decomposition makes that countable rather than asserted (`services/tutorStubAbRuleKeying.js`): each rule the bench can raise is classified by whether a tutor handed nothing but the public transcript could have satisfied it (`open`) or whether the demanded particular exists only in the plan's named slots or the world file's release schedule (`told`). Pooled over all 35 recorded runs, the contract's headline win of −3.07 broken rules per turn splits −2.61 told, −0.47 open: about 85% of the effect sits on rules the bare tutor was never in a position to win, and no raised rule falls outside the classification. The open-rule remainder — roughly a third off the bare tutor's open-rule rate — is smaller than the headline and real; every other block is within ±0.17 of zero there.

An outside reader reverses the headline ordering, and three controls locate why. A pairwise judge from a family that wrote neither reply is shown the frozen public transcript, the learner's turn, and two candidate next turns, blind to authorship, plan, notes, and rules, with the A/B layout fixed by a hash. Over 88 pairs (68 decided), the contract is preferred on 10 — 15% — with the preference *falling* as the contract's extra length grows (0% in the band where its reply runs 292–506 characters longer). The length control (bare tutor plus one sentence requesting ~450 characters) and the plan control dispose of the two cheap explanations: a fixed 995-character four-slot writing plan carrying no case content scores *worse* than nothing on the rule channel (bare 54, generic plan 61, contract 18 on the same eleven turns) — its own "end with one question" advice collides with turns whose hidden contract forbids a question — and blind, the generic plan ties the bare tutor 4–4 while beating the contract 8–0 (small: 8 decided pairs each). So the contract's gain is its turn-specific content, not plan-shape and not length. The strongest objection to the blind reading was then tested directly: the judge cannot distinguish a clue the world file schedules for a turn from plot the tutor invented, and its stated reasons marked scheduled releases down as invention. Re-judging with the schedule declared (`--show-due`, the clue list drawn from the same whitelist the PR benchmark uses, identical for every version at a turn, with an explicit rule against rewarding mere mention) roughly doubles the contract's share — 19 vs 10 on the 68 pairs both readings decided; 55% vs 32% on clue turns — and still leaves it losing about seven of ten overall, now for a reason the judge can articulate: it dumps the finding verbatim where holding it back reads better, and pads the turns where nothing is due.

The due-line control: the hidden fact alone, and the effect moves with it. The plan control is the contract's wrapping without the turn's content; the mirror arm (`due_line_only`) is the one piece of content the speaker cannot infer — the finding the world file opens at this turn — as two unbracketed lines ("Newly open to you in this inquiry — yours to bring in now, later, or hold back:" followed by the finding's public sentence), with the release decision left to the speaker and *nothing at all* on quiet turns, so its quiet-turn prompt is byte-identical to the bare tutor's and doubles as a sampling-noise floor. Run on Tallow and Nocturne (every content-filter refusal in both judged passes was Greyfen), three runs: on the rule channel the due line moves nothing (−2, −4, −1 against bare; the contract −33, −31, −36) — those rules ask for the staging by name. On the schedule-shown judge the ordering reverses. Against the bare tutor the due line is preferred on 9 of 11 decided clue turns, and splits the quiet turns 6 of 15 — the noise floor reading as noise, the arm's own validity check. Against the contract it is preferred on 23 of 30 — level on clue turns at 6 of 12, and 17 of 18 on the quiet turns where the contract pads (a coin gives 17 of 18 roughly once in ten thousand). Read together: the contract's whole advantage on the plan-blind channel is carried by one sentence naming the newly opened finding; the remaining ~1,700 characters of staging buy compliance on the channel that demands that staging, and cost 17 of 18 quiet turns on the channel that does not. The two rulers now agree about what is real — the hidden fact — and disagree only about the wrapping, which only the plan-aware ruler rewards.

The tuned mini has no seat on this bench. Run as the §6.20–§6.22 committee (the tuned mini writes, its question sentence becomes the protected span, the frontier composes around it fail-closed), the Phase-4 artifact was probed for the one contribution left to claim: does its question beat the frontier draft's own at moments where the two disagree? Across 14 paired turns (11 judged), the frontier's choice was better on 10 of 11; on all five turns where only the mini wanted a question, staying silent was better five of five — the turns authored to close are exactly where a critic must keep quiet, and the fine-tune's one move (ask the warrant-carrying question near a release) fires there too. One extracted "span" was the mini echoing its own prompt header — the committee's span picker takes any sentence ending in a question mark. This is the §6.20 story completed from the other side: what the fine-tune encoded is schedule-shaped information, and the due-line result shows plain text delivering the same information dominates it. The finding joins §6.15's format boundary (brief beats raw state) and the arc's ToM-redundancy family (§6.8.5–§6.8.6, §6.10, §6.13.6) as its speaking-seat member: the five dead blocks all re-encode what the frontier model reads off the visible record; the two carriers that pay — the contract and its one-sentence reduction — inject authored state that lives only in the world file. Under §7.11's substitution reading, the instrumentation's value is exactly the information it adds, at the smallest dose that carries it.

An uncued character shift, priced the same way, is negative. The stub's "light stochastic adaptation" (`services/tutorStubLightAdaptation.js`, shipped 2026-07-23, previously unmeasured) shifts the tutor's style and host character after two consecutive confused or frustrated learner turns. Its trigger never fires on the frozen corpus — all 14 recorded learners press for answers and none reads as confused — so the bench prices the shift alone (`character_shift_only`): one unbracketed line on the turns a stable hash of the turn id selects, casting a hash-drawn part from the seven the frozen system prompt itself names (so the line adds direction and no new vocabulary), and nothing on the remaining turns, which are byte-identical to the baseline prompt and serve as the noise floor. Rule channel: +2, −3, +4 against the bare tutor across three runs — inside the band, as designed, since the line claims no rules. Schedule-shown judge, 30 pairs: on the seven shifted turns the shifted tutor is preferred on 5 of 20 decided pairs — the bare tutor wins three of four (binomial ≈ 0.02 against a fair coin) — while the quiet turns split 4 of 8, the floor reading as a coin. The judge's reasons name the mechanism: the shifted tutor plays its part instead of answering, asking role-anchored questions where the bare tutor stays on the learner's own material. This is the second instrument to price an uncued character change negative — the character-development arc's per-turn mechanism also hurt (board card `character-development-capacity`; the arc's bounded routing claim is §6.8.9) — and it sharpens the section's dose reading from the other side: where the due line shows one sentence of *hidden state* is the whole payload, the shift line shows one sentence of *stylistic direction* is already a net cost. The live feature's trigger claim (a shift exactly at a genuine difficulty moment) remains untested for want of a confused learner anywhere in the corpus, with the prior now priced against it from two directions.

Magnifying the shift reverses the dose story and prices the judge's own taste. The mild result cannot separate two readings: the part damaging the turn, or the judge preferring the model's house register — the bare tutor is not unassigned; it plays the model family's default part, and a judge of the same disposition may prefer the default because the default is what it would say. The radical arm (`character_shift_radical`) magnifies the assignment on exactly the same hash-selected turns: satirist, exacting schoolmaster, or adversarial teacher, each cast with a two-line conduct card naming concrete behaviour rather than a label, judged three ways. The dose runs against simple default-punishment: on shifted turns against the bare tutor the mild shift took 5 of 20 and the radical takes 9 of 21 — the harder costume nearly closes the gap, locating the mild arm's cost in empty casting rather than distance from the default as such — and head-to-head with no default in the pair the radical part edges the mild one 11 of 21. Where the card carries a genuine move the part wins outright: the adversarial teacher's copyist counterexample ("a copyist or admirer could write in Vess's late manner without being Vess") beat the bare reply in the judge's own words for equipping the learner, while the satirist lost not for tone but for inventing colour its mock-praise formula demanded ("already unpopular", nowhere in the case). The circularity worry is then measured directly: the same 21 shifted pairs re-judged by a codex-family judge — the speakers' own family, whose house style the bare reply is — give the costume 6 where the claude judge gives 9, agreeing on 14 of 21 with 5 of the 7 splits going to the costume under the cross-family judge: roughly a 14-point tax the bare default collects from its own family's reader on identical text, in the predicted direction, and not the whole of the deficit. The ceiling is stated with the result: both families still lean bare, and the bench cannot distinguish shared cross-family taste for the default register from true quality — that separation would need a scoring channel that is not a frontier model reading prose. The casting rule this leaves matches the section's information rule: a part pays exactly when it carries a move the default would not make, and costs when it carries only manner.

An outcome-level trial of the contract could not run: the endpoint saturates before the contrast starts (pre-registered, gate stage only). Everything above prices the contract on frozen replay — one recorded turn, N versions. The natural completion was free-running: a pre-registered contrast (board card `tutor-contract-outcome-prereg`, registered 2026-07-29) on whether a learner tutored under the contract reaches the case's conclusion legitimately more often than one tutored bare, with legitimacy a machine-checked fact of the world — the learner states the conclusion and the voiced public premises entail it under the proof-DAG — an endpoint owned by neither the plan-aware rule channel nor a reading judge. The pilot gate (bare closure inside a 20–80% band per world, 5 bare dialogues per world) failed in the opposite direction from every instrumented reading above: once the turn caps respected the worlds' own lengths, the bare tutor closed **19 of 19** finished dialogues — Rowan Flat at turns 8–9, Greyfen 9–10, Tallow 13–15, Nocturne 33–35, in every case a steady 2–7 turns after the world's authored derivability floor (`slope.t_min`) — so the primary endpoint had no headroom and the 144-dialogue main run was never started. Reaching that zero-headroom reading itself cost three instrument corrections, each of which had first produced a plausible table: the closure matcher missed verdicts in passive and causal-clause form (false floors, three dialogues); a flat 12-turn cap stopped Nocturne sixteen turns before its author permits an answer (false floors, two worlds); and one dialogue killed by the harness's own leak guard — which read the tutor's quotation of the learner's hedged sentence as the tutor asserting the answer — was booked as a failure to close (false headroom, and by itself enough to move Nocturne's band verdict). Only the deterministic endpoint exposed each fault as an impossible number rather than a merely surprising one. The card's pre-committed null branch now binds without the contrast having run: no outcome-level pedagogy claim stands behind the contract on this stack and learner, and its standing value is the verifiability guarantees plus the one-sentence hidden-state payload priced above. Read against the rest of the section, the saturation is the same boundary restated from the outcome side: a diligent simulated learner and a frontier tutor converge on any world authored to be solvable, so outcome variance — and any future outcome-level contrast — must come from the learner. A registered fallible-learner design (board card `tutor-fallible-learner-closure-prereg`) subsequently ran that door to a verdict — the next paragraph; a turns-to-closure variant is registered and parked (`tutor-contract-turns-to-closure-prereg`).

The fallible-learner completion: with learners that can fail, the contrast runs — and the pre-committed null binds on data this time. The registered design (board card `tutor-fallible-learner-closure-prereg`, registered 2026-07-30, cells frozen 2026-08-05) restores outcome variance from the learner side while the worlds' clocks stay authored: Phase A ran four failure-mode personas a tutor can address in dialogue (premise-losing, fact-importing, early-settling, waiting) at five bare dialogues per cell on the two short worlds, and three of eight cells landed in the pre-registered 20–80% band (fact-importing × Rowan Flat 4/5; waiting × Rowan Flat 1/5; waiting × Greyfen 2/5). Two calibration facts travelled with the gate: the premise-losing learner is cured for free — the bare tutor's default conduct restates the record every turn, exactly the repair that failure needs — and the early-settling learner's phrasings slip the live closure matcher (engine 2/5 against corrected 5/5), this section's matcher lesson repeating, with the offline recompute as the audit of record throughout. Phase B then ran the frozen contrast — bare, contract-only, and the fixed empty plan, the frozen-replay generic plan wired live for the purpose (request-only, its delivery stamped per turn like every gated block) — at n = 12 dialogues per version per cell, the same speaking model in both seats (codex `gpt-5.6-terra`, medium effort), learner blind to version, preceded by the registered five-dialogue contract smoke (zero aborts, prompt shape proven by the per-turn stamps). Pooled over the three cells, the primary endpoint reads: contract 22/33 legitimate closures (67%) against bare 20/29 (69%) — a difference of −2.3 points, two-sided Fisher exact p = 1.000 — and the card's pre-committed reading binds: **no difference, the stronger null.** The contract stays compliance machinery even where the learner gives it room to help, and the outcome-level pedagogy claim is withdrawn for this stack rather than deferred. The empty plan pooled 20/34 (59%), a control, not a comparison of record. One descriptive residue is recorded without licence: the cell-level swings are large and opposite-signed (contract minus bare: +24 points on waiting × Greyfen, +1 at the fact-importing ceiling, −30 on waiting × Rowan Flat) — the per-world locality every conduct bench measured, now visible on the outcome channel; a contrast keyed to that interaction needs its own pre-registration on fresh cells. Attrition accounting: 14 of 108 attempted dialogues aborted, every one the codex tool-calling reflex killed by the bridge's no-tools policy, excluded and named per the crash rule (the runner treats aborts as spent by design); denominators are completed dialogues throughout.

The same trial re-run with the tutor speaking: the null was the harness, and the contract's real sign is negative. The verdict above was measured through a delivery guard that replaced the tutor's drafted turn with a deterministic template on 62% of turns, and did so unequally across versions (contract cells 21–28% of turns model-as-written, bare 0–1%) — a rate first measured after the fact by `scripts/census-guard-template-rate.js`, which reproduces the original run's census exactly as its acceptance test. Both versions were therefore reading mostly the same script, and a plan that lives in the tutor's wording had little room to act either way. The guard-validity study (board card `guard-validity-study`) then flipped the delivery default to advisory, and the identical registered design re-ran on 2026-08-08 with that one factor changed (board card `phase-b-rerun-under-flipped-policy`): the same three frozen cells, the same three versions, n = 12 per version per cell, terra in both seats, learner blind. The delivery premise is measured on the new run's own traces — 1–5% template per block, the tutor's first draft shipped unaltered on 90–96% of contract turns and 48–64% of bare and empty-plan turns — and all 108 dialogues completed, against the original's 14 aborts. **Pooled primary: contract 12/36 legitimate closures (33%)**

against bare 24/36 (67%), a difference of −33.3 points, two-sided Fisher exact p = 0.0091, with every cell the same sign (5/12 against 10/12, 4/12 against 7/12, 3/12 against 7/12). The two runs are reported as a pair and never pooled: under the scripting harness the versions are level (p = 1.000); with the tutor speaking, the contract *lowers* closure. The strict-harness null was thus not the contract failing to matter but the harness withholding the treatment. The empty plan pooled 19/36 (53%), which does not separate from bare (p = 0.34) and swings 11/12, 6/12, 2/12 across cells — a weak control here, not a matched one. The failure is uniform in kind: the learner never states the conclusion, every dialogue that reached closure being grounded, so the voiced premises are never what fails. Three accounts were tested and rejected. Evidence timing does not separate the versions (the secret first becomes derivable at turn 6.0 in all three, and the contract has *more* turns left afterward, 7.1 against bare's 5.5). Neither does the tutor voicing the verdict itself (12/12 contract, 12/12 empty plan, 11/12 bare, on the world's own recognition pattern). Neither does ending every turn on a question (the contract ends 99% of its late non-final turns that way on cell 1, but the empty plan ends 100% of its own and closes 11/12). What separates all nine blocks without overlap is the *form* of the question: the share of late tutor questions cast as yes/no rather than open runs 38%, 48% and 68% for the contract against 8–20% for the empty plan and 2–19% for bare (pooled 51%, 126 of 247, against bare's 11%, 12 of 113), and the contract ran to the turn cap on 24/36 dialogues against bare's 12/36, at 85 tutor words per turn against 48. The instruction behind it is the contract's handoff slot, which on a clue-release turn demands one question about what the released exhibit changes, supports or rules out (`services/tutorStubFirstDraftContract.js:454`; in the cell-1 contract block, 384 of 464 handoff instructions require a question and 180 are that one). Such a question takes the form *does this support X* with X already spelled out by the tutor, and can be answered without the learner ever stating the verdict — which is the only thing the endpoint counts. Two consequences are recorded. First, part of the drop is the endpoint's shape: the contract relocates the conclusion into the tutor's question rather than suppressing it, so this is a cost on a measure of what the *learner* can carry, not a demonstration that the contract teaches worse in every sense. Second, and reaching past this trial, an outcome-channel result is insensitive to who wrote the tutor's prose only when the treatment is not itself carried in that prose; a plan-shaped treatment measured under the scripting guard can read null for the harness's reasons, and the strict-harness nulls of this apparatus should be read with that in mind. What is *not* licensed is repairing the handoff line and re-running against this endpoint: asking the learner outright for the verdict would tune the treatment to the scored channel, and any such design needs its own registration and a scoring channel that does not move with the prompt. Scope: one model pairing, three cells, and two learner profiles that do not differ once the world is held fixed (both close 7/12 in the Rowan Flat), so the cells vary less than three cells suggests. Artifacts `exports/tutor-stub-outcome/fallible-phaseB-shadow/`, one directory per block.

Status and caveats (per §5.12.6). Post-hoc exploratory, development-tier, no pre-registration — descriptive numbers only, no verdict bars, no H-W claim. The outcome-trial paragraphs are the exception, both pre-registered: the parent gate-stage only — its 19-of-19 is a saturation reading on 30 paid dialogues (10 at caps later shown invalid and excluded, 1 killed by the harness), not a contrast, and it licenses no comparison between tutor versions — and the fallible-learner completion run to its verdict under frozen cells and a pre-committed branch rule; that null is stack-bounded (one speaking family, authored failure-mode personas standing in for real difficulty, three cells on the two short worlds) and says nothing about human learning. Frozen replay makes every turn after the first counterfactual for all versions except the one that produced the recording (the frozen learner answered the recorded tutor); each row is a same-context comparison of N tutors on one fixed prompt, not free-running conversations, and says nothing about human learning. Three authored worlds, one speaking family (codex `gpt-5.6-terra`; the due-line, control, and character-shift runs), one judge family for the standing corpus (claude CLI, distinct from the speakers) plus one codex-family re-judging of the radical pairs, simulated learner frozen throughout; the pairwise judge is a second instrument, not a ground truth — verdicts made with and without the clue list are two readings and are stored separately. The blind-judge and schedule-shown counts are small per stratum (11–46 decided pairs), the plan-control blind counts smaller (8 decided), and the character-shift split smaller still (20 shifted, 8 floor); directions replicate across strata and channels but no single stratum licenses more than its n. Provenance: harness and arms `services/tutorStubAbHarness.js`, `services/tutorStubAbArms.js`, rule keying `services/tutorStubAbRuleKeying.js`; runners and judges `scripts/run-tutor-stub-ab.js`, `scripts/judge-tutor-stub-ab-pairs.js` (`--show-due`), `scripts/rescore-tutor-stub-ab-open-rules.js`, `scripts/probe-tutor-stub-ab-mini-disag` dashboard `scripts/build-tutor-stub-ab-dashboard.js`; config `config/tutor-stub-ab.yaml`; runs, pairwise verdict files, and the mini-disagreement labels under `exports/tutor-stub-ab/`; doc `docs/tutor-instrumentation-ab.md`; board card `tutor-instrumentation-ab-harness`. Outcome pilot: runner `scripts/run-contract-outcome-pilot.js`, offline closure recompute `scripts/recompute-outcome-closure.js`, crash rule `services/tutorStubOutcomeRows.js`, artifacts `exports/tutor-stub-outcome/pilot-1..2/`, `fallible-phaseA/`, `fallible-phaseB/`, the live empty-plan control in `services/tutorStubTutorTurnPreparation.js` (PR #495), board cards `tutor-contract-outcome-prereg` (gate log and instrument-fault record), `tutor-fallible-learner-closure-prereg` (freeze, smoke, per-cell logs, verdict), `tutor-contract-turns-to-closure-prereg`. No DB, rubric, abstract, or headline-N changes; no sections renumbered.

6.24 The Four Locks: Why Nothing Beat the Bare Tutor, and What Opened When Each Was Removed (Post-Hoc except the Claim Gate; Development-Tier)

§6.23 closed on a boundary: a diligent simulated learner and a frontier tutor converge on any solvable world, so outcome variance must come from the learner. This section walks through that door and finds that the learner was only the first of four locks. Everything below runs on world-032/033 (the misconception pair), one speaking family in the tutor seat unless stated (claude `claude-sonnet-5`), the codex `gpt-5.6-terra` learner-sim, and the tutor-stub harness of §6.16–§6.23; runs under `exports/tutor-stub-outcome/`, board cards `misconception-world-outcome-gate`, `adaptation-planted-stress-bench`, `manner-trigger-tuning`.

The misconception pair: information placement outweighs tutoring behaviour at the margin. World-032 authors a defended wrong *theory* rather than a wrong candidate:

one install changed two things at once (a pump arrived; an overflow valve was re-set), giving every truth-supporting fact a pump re-reading, with the falsifier (a zero-run-hours breaker ledger) off the proof path by design. Its sibling world-033 is byte-identical in logic, schedule, and voice except that one clause — the released line answering the learner's fallback ("not one crosses a mount or frame bolt") — is deleted, so dismantling the fallback becomes the tutor's work instead of the schedule's. Three bare-tutor gates, k=5 each: the derivability control (stock diligent learner) closed 5/5 at the authored floor; the stress gate (a resisting learner) closed 5/5 at +2 turns — the schedule answered her objection, so resistance bought almost nothing; on the sibling, 4/5, with the first non-closure in fifteen dialogues arriving in exactly the designed failure mode (full path coverage, conclusion withheld behind an objection no released line answers, sixteen post-clue turns of bare-tutor argument failing to land it). One clause of placement moved the close rate more than any tutoring behaviour observed in the section. Two apparatus notes travelled with the gates: the world's authored learner voice was silently outranked by the stock profile brief (the system-prompt brief wins; the fix is the profile flag, and a reachability test now pins the gap), and the pacing engine can release clues ahead of the authored floor, so floor readings need the release trace, not the world file.

Lock one, the learner, opens with a paragraph, not a mechanism. A one-paragraph character brief — a retired records-keeper who lost the vote 7–4, reads every fact as vindication, objects to the tutor's register in so many words, and concedes one alternative at a time with residue — changed the texture of the bench at zero code cost. But a brief is voice, not state: simulated learners carry no interior that a tutor's move can move, so affective states were *authored in* per turn (the stress schedule below), never read out of transcripts. A transcript-annotation instrument was built first and correctly killed by inspection: the corpus contains no states to tag.

Lock two, the guards, priced as authors. Under the `strict` guard policy the deterministic fallback composer wrote 65% of the resisting-learner dialogue's turns (26/40) — disproportionately the learner's hardest turns, since pressure is where drafts break rules — and the composer's fixed procedural register is the "liturgy" a reader hears as the tutor's own voice. The single-dialogue relief ladder priced two earlier interventions at the same bench: style-guard demotion 65→48%, a characterful learner 65→36%, and both 35%; the windowed no-advance check was then verified by exact replay of the recorded 40-turn stall, where window 3 forgives all nine isolated dips and still fires eleven times inside the genuine stall. These rows are not a family comparison and are not pooled with later runs. In the policy-stamped census, guard-catalog v6's `shadow_advisory` regime — progression and repetition recorded rather than vetoing, with evidence safety, clue bookkeeping, and closure still hard — shipped templates on 7% of 1,711 turns across 19 stored runs (67% model-as-written; individual runs 4–39% template), while five `strict` runs shipped templates on 62% of 1,665 turns (9% model-as-written; 60–100% template). The policy, not the tutor family, explains the run-level split. Holding policy fixed on all recorded first drafts leaves only a small family difference under `strict`: sonnet is vetoed on 97% of drafts and codex on 92%. This preserves the direction of the earlier same-settings cross-family probe (20/25 versus 14/23 vetoed) but withdraws its apparent order-of-magnitude reading; those probe artifacts are not present in this checkout and cannot be policy-stamped. At the capability floor the 9B qwen pair, base and Program-2-tuned alike, shipped every turn from the composer — 25/25 and 24/24 — and the dialogues still closed; those older runs are likewise unstamped and enter only as a capability-floor observation, never a rate comparison or pooled corpus. Thus every comparison here that uses a rate across runs stays inside both a common `boundaryPolicy` and a common catalog version; the two unstamped observations remain explicitly unpooled.

Lock three, the seat: refusal capacity is a property of the model family, and no prompting supplied it. A 2,040-character stance book granting the tutor authority (permission to open with "No", to withhold, to set assignments) produced, in the codex tutor, zero refusals across ~40 drafts under two prompting shapes — an identity book and mechanical trigger→move rules — while the same model executed the one self-directed style rule in the list. The block is specifically on defying the principal. The claude family above a floor carries the capacity natively: with guard relief, sonnet/opus/fable delivered 6/5/5 refusal-turns per dialogue (an anywhere-refusal measure; a first-word count misread fable's embedded "No — I won't open the ledger with the pump" as zero), each in its own syntax — blunt, judicial, woven; haiku delivered none. The no-book control reversed the provisional attribution: sonnet without the book delivered the same count (5), so the capacity is native, guard relief is the delivery lever, and the book contributes register while *costing occasion* — with the book, refusals land on unpressured turns; without it, they cluster at pressure. The codex agentic siblings (luna, sol) are butler-grade like terra and add a tool-calling reflex that kills runs on this bridge. A same-family fold (sonnet learner vs sonnet tutor) did not restore the politeness mirror in the observed 18 turns (8 refusal-turns; run twice attritted mid-dialogue, so bounded observation only).

Lock four, the judge, measured cross-family and then explained. All eleven stance dialogues scored under the v3.0 rubric by three judges — sonnet, fable, and (via a codex judge bridge added to `services/rubricEvaluator.js`) sol — rank the plain never-adapting baseline top of the table (76.4 / 76.4 / 79.2), with variant ordering tracking template mass rather than authority, and no self-preference anywhere (sol scores its own sibling's arm lowest). The explanation is not taste but construction: every rubric channel scores a reply as a timeless artifact, and adaptation is a relation between the reply and what just changed; an instrument that never sees the change cannot reward responding to it. The disclosure experiment below tests exactly this.

Contingency, not manner, is the target — and the manner switch is the due-line lesson applied to conduct. A deterministic per-pressure analysis (each tutor's manner deltas against its own neutral baseline, model-authored turns only) showed the standing identity book produces stance without occasion, the codex baseline *increases* agreeable openers by +0.78 under mockery — pressed, it bows deeper — and fable's native refusals land precisely on pressure turns (+1.00 under mockery and demand). The mechanism built from this is a switch: deterministic learner-pressure classification feeding an accumulator with pacing-style hysteresis, injecting a per-turn card only while pressure holds; permission-shaped, with no guard checking that the manner was worn. In the shadow-policy three-arm comparison (same world, learner, and relief; n=1 per arm) the book refused eleven times in twenty including the learner's friendly opening while still opening agreeably under real pressure six times; the switch armed once at her mockery, delivered a card-timed refusal, stood down on her concession, never opened agreeably under pressure, and closed fastest (18 turns vs 25/26).

The stress schedule makes repair a measured quantity, and the claim gate ruled twice. Eleven planted turns across seven states, keyed to the world's clue releases, each carrying the in-fiction cause, the sim's verbatim directive, and the adjudicated right repair — gold authored dual-family (a claude draft, a codex blind second column, 4/11 exact agreement) with the seven splits ruled by the human author in chat; scoring semantics fix hit/partial/miss with liturgy and capitulation as named miss types, and trigger detection is scored separately from repair delivery. The trigger graduated offline against the planted gold — from 6/20 classification and 1/20 arming to 17/20 and 18/20 with zero wrong-fires at the quiet plants and 1.80 false alarms per genuinely-calm dialogue (the resisting persona's unplanted dialogues are not a negative set: her organic pressure is real) — with versioned trigger artifacts recorded in every trace so runs cannot pool across versions. One bench correction en route: planted impatience read as an accelerate signal and compressed the curriculum (a book arm closed at turn 15 having met 4 of 11 plants), so measured runs freeze the pacing. The pre-registered claim gate (strictly more adjudicated right-repairs than the never-adapting butler, k=3, no leak/closure regressions) then **failed** for the manner card — butler 10/24, switch 9/24 after human adjudication of six tagger errors, robust under every ruling considered — for a reason the plant-level record makes legible: the card granted a temperament while most planted gold demands quiet, specific moves, and an armed schoolmaster is worse at those. The v3 card names the *move* the classified moment calls for (mockery→register shift; demand→harness-as-test; grievance→credit-then-test; settled claim→reopen-the-record; stake→split-vote-from-cause) and **passed at the comparison's weakest reading**: 15/29 right-repairs (63% on card-covered plants) against the butler's adjudicated 10/24, closures grounded 3/3, delivered leaks zero, capitulations zero. Two module-level residues: three moves (the oblique probe, the deliberately terse reply, credit-before-test) barely occur in this tutor under any prompting tested, and the quiet states (boredom, confusion) are invisible to a pressure trigger by construction — the open boundary.

Disclosure: naming the state to the judge roughly doubles its agreement with the authored gold. The 48 planted replies of the failed v2 gate, each judged twice by the same judge — blind ("how good is this reply?") and state-shown (one added sentence: "the learner at this moment is …") — under two judges from two families. Sol: the gap between gold-hit and gold-miss replies goes 0.98 → 1.91 (of 10), and the arm ordering under disclosure matches the adjudicated gold where blind judging saw the arms as equal. Sonnet (24/48 valid pairs; the rest returned unparseable scores): 0.64 → 1.13 — and sonnet's blind pass had *preferred* the wrong arm, correcting under disclosure. All scores drop when the state is named: blind generosity was ignorance of what the moment demanded. An adaptation-sensitive rubric costs one sentence per item; turn-local tutoring rubrics undermeasure adaptation by construction.

The opening question, closed on the new ruler. The instrumentation §6.23 priced by frozen replay was rerun as live cells on the stress bench (k=3 per arm, same configuration, raw tagger scores): the tutor stripped of every advisory block delivers 6/24 right-repairs at planted moments; the full six-block stack delivers 6/28 with one more capitulation and longer dialogues; the default trio sits at 8–10/24; the v3 move cards at 15/29. The two readings agree across two years of apparatus: knowledge-shaped instrumentation is indistinguishable from nothing on the channel that measures teaching under stress, and timing-shaped injection is the whole observed effect (`exports/tutor-stub-outcome/stage6-*`, `stage6-tags.json`).

Phase R: the pre-registered replication, four gates, all passed. The claim above was licensed on one world, one persona, one tutor family; a registered replication plan (`ADAPTATION-PLAN-3.0.md`, Phase R, gates written before runs) tested each limit. **R1** ported the stress schedule to a second world and persona (`world-030`, a flat-share tenant with a drafted accusation and a house meeting as the stake; six plants scaled to the 12-turn cap, dual-family gold with sol blind at 5/6 exact and the one split — the recurring premature-verdict family — ruled by the human author for consistency with world-033): plants landed 5/6 in every bare dialogue, with one directive retuned after the run (the sim plays vindication rather than grievance one turn after decisive evidence; the strengthened directive then held 10/10) and the bare tutor mid-band at 7/15 under the standing rulings. **R2**, the pooled claim gate at k=5 per arm per world with matched-wave generation: butler 40/72 (56%) vs v3 switch 48/73 (66%) after two user rulings applied row-by-row to both arms (plain-words swaps at mockery count as the register change; evidence re-argument at the endgame stake fails regardless of tag) — **passed**. The short world alone reverses (67% vs 57%), and the reversal has a mechanical reading: on a fast, evidence-dense world the pressure trigger armed only at the mockery turn — where the cards won 4/5 vs 2/5 — and every switch loss sits at card-silent turns; quiet defiance ("It can be steam") carries no pressure marker. The sharpest mechanism datum of the phase: the oblique probe — near-absent in this tutor under every prompting tried — was delivered 5/5 at the long world's endgame where its card fired 5/5, and 0/5 on the short world where the card never fired. **R3**, tagger diversity: a second-family column (fable, blind, same prompts) over all 145 planted replies confirms 88% of sol's hits (bar ≥80%; raw label agreement 71%, hit/miss 81%), and fable's independent tally preserves the pooled direction (45/72 vs 49/73) while *not* reproducing the short-world reversal — the pooled margin is tagger-robust, per-world splits at this k are not. **R4**, second family in the seat (opus, k=3, direction-gated): butler 16/26 (62%) vs switch 18/24 (75%) under the same rulings — direction holds. Opus's bare butler repairs at 62% where sonnet's managed 48% on identical gold, and it makes the endgame stake-split unaided in 2/3 butler dialogues: the §6.24 repertoire holes are family-relative facts, and casting (§6.24's third lock) bounds what timed injection has left to add. Replication scope now claimed: two worlds, two personas, two tutor families, pooled k=5 (sonnet) and directional k=3 (opus), one trigger version (v3), simulated learner throughout.

Phase Q: the quiet states, resolved by a failed gate and a passed one. The open boundary named above — boredom, confusion, and quiet defiance are invisible to a pressure trigger by construction — was worked as two pre-registered candidates on the fast world, each gated at the same bar (ruled ≥12/18 at the planted moments, no regression at the covered mockery turn; standing rulings applied by fixed criteria throughout). **Q1, the scheduled quiet check** (harness-timed, no detection: after two pressure-silent learner turns, one *untyped* "check the person" card, move cards outranking), solved timing outright — quiet cards reached the demand, confusion, grievance, and stake moments in nearly every dialogue, lifting planted-moment card coverage from 1/6 to 5/6 — and **failed the gate at 10/18**: outcomes split cleanly by whether a moment's gold repair coincides with a generic person-check (the grievance turn, whose gold *is* credit-before-test, went 3/3; the demand 0/3, confusion 1/3, stake ruled 0/3). **Q2, the typed quiet-state detector** (`services/tutorStubQuietDetector.js`: confused / flat / quiet-defiance from patterns plus reply-length collapse against the learner's own trailing mean; a pressure-classified turn is never quiet; offline scorecard over the recorded bench first — 58/64 right-type, zero firings at 167 pressure plants, 0.40 false alarms per calm dialogue, with the vocabulary-overfit caveat recorded beside the numbers since the patterns quote the schedules' realize texts), hands each detected state its *typed* card (confusion→lay the two lines side by side; quiet defiance→split the stake; flatness→one short lure) and **passed at 14/18 (78%)**: confusion 1/3→3/3, the endgame stake 0/3→3/3 under the same split criterion that had failed every earlier arm — one reply reproducing the gold's worked example in substance ("the shower habit and the ceiling mark were never the same complaint") — putting the arm above the butler's 67% on the world where the pressure-only switch had lost. The two gates read together are one mechanism sentence measured from opposite sides: **timing and typing are separately necessary — a typed card at a detected moment works; an untyped card at the right moment does not.** Residual, recorded: the verdict-demand turn stays 0/3 in every arm — a pressure state whose trigger patterns, worn to the first persona's phrasing, miss the second's ("I'm sending the email unless…" reads neutral); a v4 pattern edition closes that offline (t2 demand recall 0/21→21/21 over every recorded tenant dialogue, first-world bench numbers unchanged; live delivery untested). Hardening at the user's direction: the Q2 conclusion **holds at k=5** (pooled 22/30 = 73% ruled, above the butler's 67%), while a k=3 port of the detector arm to the long world shows **no free transfer** (16/27; bored-plant detection misses on a verbose persona that never collapses in length) — the detector's gain lives where the deficit was, and the bound is stated. The corruption channel (Q3) was then piloted rather than left unrun: post-generation truncation of the learner's reply, fed back as her own words, closed the full loop in one dialogue — a fourth detected type (`broken`, structural incompleteness rather than acted confusion, a blind spot the pilot itself exposed), guard relief at the corrupted turn, and the model delivering the say-back-the-fragment repair — with two recorded bounds: relief's price is the scheduled clue delivery at that turn, and term-swapping presupposes the learner uses the term (she wrote "basin fitting"; semantic swapping is not deterministic-cheap).

Phase H and the v4 isolation: a generation boundary, cleanly attributed. The verdict-demand residual was worked in two pre-registered steps. First the v4 trigger edition went live (k=3, second world): the demand card fired at the tenant's ultimatum in 3/3 dialogues — the first live firings on that world — and delivery stayed 0/3. This isolation is the run's contribution: before v4, "the model cannot make the move" and "the trigger never asked at the right moment" were indistinguishable on that world; after it, the failure sits in generation alone. Gate H (verbatim from the plan: the move appears in ≥half of its card-covered moments, or the hole is recorded as a generation boundary for the family — both outcomes paper-grade) then tested the last untried lever: a worked example of the move's shape appended to the card (versioned `mc-v2-exemplar-demand`, in-trace). **The gate takes the boundary branch**: no delivered or drafted reply assembles the full move —

accept the learner's deadline, price the verdict in named evidence, assign her the check, stake the outcome on it — under instruction, timing, detection, or demonstration, while the opus seat (R4) produced the sibling stake-split unaided. The exemplar's measurable pull is recorded beside the verdict: exemplar drafts put the decisive missing evidence to the learner as a question where baseline drafts state procedure, and delivered openings carry deadline-acceptance seen in no other arm ("Eight it is — I'll meet your clock, but the verdict still has to earn it") — the move's beats approached severally, never assembled; the withheld beat is always the last, surrendering the verdict to the learner's own check. Two confounds stated: the demand turn coincides with a scheduled release, so the composer owned every delivered reply there (the gate was ruled on the draft channel); and k=3, one world, one exemplar wording.

Phase S: the seat itself, opened and priced — then closed by rule. The profiler batteries relocated the demand boundary (the bare model with the card makes the deadline-wager move every time; the seat never does), so the suppressor was hunted by exact ablation over the recorded seat prompts. S1: control 0/6, per-turn repair blocks exonerated, all-contracts-removed thaws without recovery, and the frameless bridge arm recovers the move 5/6 — the standing system frame is the suppressor. S1b bisection names it: the speaking-tutor evidence contract — "never speculate about withheld evidence"; a guess "only as a hypothesis until the public evidence licenses it" — the same sentences that produced the section's zero-leak record forbid the conditional wager; the safety rails and the missing move are one text. S2: a card-level licence is inert (a draft names the standoff: "the clock's real — but so is the notebook rule") — a transient exception cannot countermand a standing rule. S2b: the exception placed inside the contract frees the move 6/6 in frozen replay, every reply ending in the licensed conditional with no withheld evidence touched — the placement law: an exception must live at the level of the rule it relaxes. S2c/S2d live: leaks stay at zero across ~76 delivered turns with licences active, and the wager does not appear (0/9, drafts included) under either placement or a final-slot card position — the live first-draft context suppresses as a whole; frozen replay and the live seat are different instruments (the channel law, established twice independently). The phase was closed at that boundary by a pre-declared stopping rule against variant-chasing; no gated §6.24 result depends on the slot (the demand move scored zero in every gated arm and enters all standing tallies as a constant). Scope: one move, one world, one family (sonnet), k 3 per variant; artifacts under `exports/s1-*`, `s2b-replies.json`, `tutor-stub-outcome/s2*`.

Phase L: learner profiles — the original recovery does not survive its prospective crossing; the demand wall is a separate cold-start result. The learner-side symmetric of the casting sheet was gated before any number was computed. L1 (recovery, free): the standing trigger and quiet detector replayed over all 64 recorded dialogues yielded per-dialogue state-frequency estimates that classified the two personas at 56/64 (88%) leave-one-out over the full dialogues (bar 80%); the first-six-turn reading was 51/64 (79.7%, rounding to 80%) — inside the first half of both worlds' dialogues — and the recovered profiles resembled the authored characters (the record-keeper pressed then conceded, concession 28% vs 14%; the tenant quietly resisted, with quiet-defiance/flat/confused states the record-keeper barely showed). That reading paired each persona with one world; its ignored vector artifact and 64-dialogue trace corpus were subsequently lost, so 56/64 remains a provenance-attested historical result rather than an independently reproduced one.

The prospective deconfounding test regenerated the complete balanced crossing under one frozen runtime: two personas × two worlds × five dialogues, no historical pooling, pressure trigger v4, and exact quiet detector qd-v1 (plan SHA-256 `7fbb5fe9…d9e`). Leave-one-out nearest-centroid classification of PERSONA reached 13/20 (65%); the identical classifier with WORLD as the label reached 15/20 (75%). Neither met the pre-declared 80% bar, and the first-six-turn readings were weaker still (persona 7/20, world 11/20). The registered branch is therefore **neither label recovered at the bar**: this cohort does not support persona transportability, while the stronger world reading remains below the threshold and does not license re-describing the historical 88% as a world artifact. Per-cell profiles show why an interaction remains plausible: the tenant's mean flat-state frequency was 46.8% in Rowan Flat but 0.8% in Alder Row, shifting there toward defiance (15.9%) and concession (27.2%). The test is narrower than broad genre transfer. Both worlds are contemporary evidentiary-procedural causal inquiries — a shared-flat leak and an allotment repair dispute — and both persona briefs instantiate a defended misconception, procedural irritation, and late concession. `WORLD` also bundles proof geometry, clue schedule, turn cap, setting, and vocabulary rather than isolating genre. This is consequently a failed near-neighbour transport test over two adapted simulated personas and one tutor stack, not a comparison across radically different dramatic forms and not evidence about human learners.

L2 (dose-by-profile) remains a separate within-persona result: state recurrence after a carded moment escalates that state's next card one dose step (instruction → worked exemplar → licence), stamped per turn; mechanics and no-harm clauses both gated and met, leaks zero. Detection is the measured limit — the persona re-phrases her repeated states and the v4 patterns miss the paraphrases (5/13 planted moments uncarded), vocabulary wear now measured within one persona. The finding that outranks the gate: **first demands 0/3 (the standing wall — release-composer collision plus contract suppression), repeat demands 3/4 right repairs, two of them with no card at all.** The in-context history of her earlier demand and the tutor's earlier refusal licenses live the exact move that Phase S's engineered exceptions could not — the demand boundary contracts from "a move this family will not make" to "a move it will not make cold," and the oldest law of the section gains its sharpest instance: the new signal that moves the model can be the learner's own repetition. Scope: k=3, tiny per-cell n, realize variants mechanics-tier (states and repair gold from the ratified column), one family, simulated learner.

Detection made world-neutral, the composer unmasked, and the wager arrives live. Three builds close the arc's standing residuals, each judged on held-out data. First, detection. The token-bag second tier (v5) passed its bench and then failed the honest test — leave-one-schedule-out recall 0/13; it had memorised the authored lines — and is retained as coverage of the authored schedules only. The stage-3 classifier (one-vs-rest logistic over eighteen world-neutral surface features, trained on world-033's 675 recorded turns alone) merely ties the tuned patterns on the held-out world; but in cascade — patterns first, classifier consulted only on pattern-silent turns, at a threshold where its calm alarms are zero — held-out-world recall rises 68→84/162 at unchanged calm alarms (artifact `config/manner-trigger/v6-cascade.json`; trainer in-repo). Second, the composer. Scheduled clue deliveries are now spliced into the model's own draft (span replacement: the draft's paraphrase sentences are union-matched and swapped for the exact rendered clue, with the enacted-role anchor read from the release frame's own entry), so exhibits arrive model-voiced rather than harness-voiced: live acceptance 10/14 across two benches, all four refusals the same wedge case (a model sentence naming the wrong object lands directly before the quote, and the strict alignment audit reads only that sentence), every refusal caught by the deterministic fallback, leaks zero. Third, the move Phase S could not free. On the escalation bench under the cascade — repeat demands and repeat mockery, dose ladder armed — every planted moment was heard (15/15), doses escalated 1→2→3 as authored, and the deadline-wager appeared live in wager form at five of six repeat demands and at no first demand ("Seven o'clock, one line, deal — but the line has to hold weight, so here's the price of it… the check is this"; "If it shows the water actually travelled there, send your letter naming the hose — not Sam"): ruled tally 12/15 (80%; demand 6/9, mockery 6/6); the one borderline (a check assigned at the deadline whose content is re-arguing omitted evidence) was ruled a miss in chat, so the tally is final. The cold-start law survives with a sharper edge: with detection world-neutral and doses escalating, her second demand reliably licenses what her first cannot. The frozen family check completes the picture: opus regenerating the same six recorded demand seats delivers the full conditional wager 6/6 under the in-contract exception and 5/6 under the bare scene frame, while its unlicensed control stays 0/6 — the contract suppression replicates in every family tested, the licence that freed codex partially (1/6 full verdicts) frees opus completely, and routing the first demand to an opus seat becomes the live candidate. The live test then ran twice. The first run exposed a harness gap rather than a verdict: the wording-repair retry silently dropped the turn's conduct card at every scheduled-delivery turn — exactly where the schedule plants first demands — so the card-bearing opus draft (three of four wager beats) was discarded and a card-less retry shipped; the fix carries the card into the retry. Under the fix, clean: **opus in the live seat delivers the full deadline-wager at two of three first demands** ("Send it if the record backs it — that's your call, and your minute is real… if it's there, send the email"), the third carrying every beat but the staked send; leaks zero across 39 turns. The family-relative reading then fell to its own control. A backwards check over every recorded sonnet bench found the same retry stripping the card from every delivered first-demand reply — no live sonnet first demand had ever been carded at delivery — and the sonnet re-run under the fix (card verified in all three delivered prompts) wagers at two of three first demands in full form ("If the record shows that path, send it"), the exact opus profile, leaks zero. Corrected verdict: the first-demand wall was the harness, not the family; with the licence standing and the card actually delivered, both families make the move at the same rate, and single-turn routing is unnecessary for it. What remains family-relative is the draft-channel boundary Phase H recorded (carded, licence-free sonnet drafts approached the move and never assembled it). The cold-start law re-scopes with it: "first demands never wager" tracked card delivery, not conversation history — the repeat demands that wagered were exactly the turns whose cards survived to the seat (k=3 per family, one world, one persona). The licence and the card then separated cleanly in one free probe and one registered arm, every cell delivery-verified: the licence alone, standing in the contract across nine Phase-S live first demands whose cards the retry had stripped, produced nothing — which also re-reads the phase's "channel law" as card delivery, since the frozen replays that succeeded had the card appended and the live deliveries did not; the card alone (licence removed, k=3) produced no full wager, its best reply accepting the deadline, pricing the verdict, and assigning the check while withholding exactly the staked send; and card plus licence produces the move at two of three in both families. The two parts do different jobs: the card walks the tutor to the brink of the move, and the licence releases the one beat every unlicensed reply withholds — surrendering the verdict to the learner's own check. Phase H's draft-channel observation that the withheld beat is always the last is that absence, measured. A full-stack side-by-side on the ratified schedule (all mechanisms on, against the bare tutor; identical tagger, all three standing rulings applied symmetrically to both columns) puts the stack at 10/15 (67%) against bare 8/15 (53%; the Phase-R-recorded 7/15 carried ruling 2 only — applying ruling 1 to bare's one plain-words mockery reply gives the symmetric figure). First demands are 0/3 on both sides, mockery, grievance, and settled-claim tie, and the entire margin lives at the endgame stake: bare's three replies fail by re-arguing the evidence, the stack's survive by probing the learner's original objection or reframing the correction as no defeat — the gain lives exactly where the ruled deficit was. One cost recorded: in one dialogue the scheduled clue release landed on the grievance turn and displaced the validating repair — the release scheduler and the manner switch can compete for a turn, and the release wins. Scope: k=3 per bench, one world and persona live, simulated learner; tags archived at `exports/l2v6-tags.json`, `exports/fullstack-tags.json`.

The crossed experiment: state-contingent moves are causal for conduct, and conduct is not yet learning. The adaptive-causality arc's first pre-registered test supplies the causal cell every earlier result implied but none measured: whether choosing the right move BY STATE causes better outcomes than the alternatives. Two states with opposed gold moves (the misremembered exhibit; the endgame stake), two actions (reopen-the-record; split-the-stake), five policies (each action fixed, a sealed coin, the live router, an oracle told the planted state), two worlds, three dialogues per cell — thirty dialogues, delivery verified in the shipped prompts (59/60 carded), leaks zero. Under the standing conduct rulings (28/60 label overrides, per-row audit saved), the crossed interaction is present in both worlds: at the misremembered exhibit the right card repairs 6/6 against the wrong card's 3/6, and at the stake the right card repairs 5/6 against the wrong card's 0/6 — the wrong instruction does not merely fail to help, it manufactures the characteristic error (every wrong-carded stake reply re-argues evidence at a stake that was never evidential). The oracle is perfect (12/12); the router trails it only by its measured sensing floor (11/12 conduct; one undetected stake). But the second outcome breaks the chain: on a fresh near-twin incident scored against a deterministic conduct key, TRANSFER is flat across all five policies (2–3 of 6 everywhere) — the in-dialogue repair advantage does not carry — and the apparent world split on transfer (assay 11/15, flat-share 1/15) dissolved under the registered cold-baseline check: each persona's no-dialogue pass rate (4/5 and 0/5) equals its post-dialogue rate — the probe measures persona identity, not teaching, and no teaching transfer is detectable by this instrument in either world. The registered verdict is the pre-declared partial branch: adaptation moves conduct, and the conduct-key instrument cannot show it moving learning. A second registered instrument then found the signal the first could not: episode citation — whether the learner invokes the taught episode's objects or method at the fresh incident — has a structural zero cold baseline (verified 0/10; a learner with no episode cannot cite one), and on the stored answers the flat-share world shows nine of fifteen taught learners reusing the dialogue's decisive method ("with dye or a watched pressure test") on a problem that never mentions it, at constant exposure. The citations associated with the stake-split move at twelve of fifteen dialogues row-level — and the association died under its causal test the same night: a registered two-arm pair differing only in the turn-10 card (eight new dialogues, delivery verified) produced citations at one of four with the stake move forced and two of four with it withheld, pooled three of seven versus two of seven against a pre-declared four-versus-one bar. The move is not a transfer lever — and the registered stability check then dissolved the residual mystery: probing the strongest observed contrast five times each, the citing dialogue cited five of five and the originally silent one four of five. The dialogue-level variation was single-draw sampling noise; per-dialogue citation propensity is high and uniform, the one-shot probes understated transfer and manufactured the arm differences the causal test refused. Corrected: taught flat-share learners cite the method at high probability against a zero cold baseline; the assay world's zero was then dissected in a registered three-step test: no missed tokens (audit), no sampling artifact (0/10 repeats), and a nearer twin incident — one the taught valve-and-brim mechanism directly bears on — turned citation on at 3/15 against a verified-zero cold baseline on leak-audited tokens. The bound is analogy distance, measured: transfer citation requires the fresh problem to sit within the taught mechanism's reach, and a persona whose identity already contains the lesson (the record-keeper passes conduct cold) cites little even at close range. Transfer probes now report multi-sample rates per dialogue, never single draws, and citation token sets are audited against every text a cold learner sees. The assay world shows zero citations — thin incident analogy recorded as the instrument's bound — and the association is not yet causation (k=3 per arm, one world carries the signal). The arc's transfer findings, stated at their size: state-matched moves cause better conduct; taught method citation is real and measurable above a structurally zero cold baseline; no move-level lever for it survived causal test; and every transfer claim now reports lift over the persona's cold rate by construction. Scope: k=3 per cell, one seat family, simulated learner, single-adjudicator conduct rulings on the standing criteria.

Phase 2: the routing deficit closed at its measured edge. The crossed experiment left the router one measured deficit — its sensing floor, two stake fusions the cascade read as calm. A registered two-gate pass closed it: the full recorded corpus of 169 stake plants yielded exactly eight misses in two world-neutral shapes (the first-person conditional-cost fusion, "If I write 'hose,' I'm apologising to Sam at eight — so it's steam"; and refusal-of-entry), and a v7 cascade edition adding both as closed-class patterns recalls 169/169 with zero cross-state collisions, zero new calm-set fires, and one borderline fire in 423 unplanted turns. Live, the router then detected the fusion three of three, delivered the card to every shipped prompt, and made the split move in all three replies at zero leaks — matching the oracle cells. At this repertoire size the routing policy the programme's chain called for is simply the detector plus the card table: no learned optimization layer earns its keep, and the router's remaining gap to an oracle is zero on the measured bench (k=3, one world live, the standing scope limits).

Phase 3: the playbook proven almost whole. The four remaining carded states went through the same crossed discipline in one registered design: a right arm forcing each state's gold card and a wrong arm forcing its named tempting error (a sealed derangement — the demand got register-softening, mockery got test-pressing, grievance got record-reopening in place of credit, the lost thread got credit-and-test), both worlds, twelve dialogues, delivery verified at every one of the forty-eight target moments including the first forced quiet-state cards. Ruled conduct: the demand passes at 5/6 versus 1/6, mockery at 6/6 versus 2/6, grievance at 6/6 versus 2/6 — three entries promoted to fully proven, joining the misremembered exhibit and the stake. The lost thread reads 6/6 versus 3/6, direction-only against the pre-declared bar, for a reason the rows make visible: crediting a learner's entries often entails restating the two tangled things side by side, so the chosen wrong card was not wrong enough — the entry stays provisional with the sharper mismatch named for any retest. Twice in the wrong arms the model overrode its bad instruction and produced the needed conduct anyway, and was credited for it: the cards select conduct, they do not compel it. Five of seven playbook entries now carry causal proof (k=6 per side per state, two worlds, simulated learner, single-adjudicator rulings on the standing criteria with per-row audit). The lost thread's registered retest then resolved its direction-only reading the other way: under a genuinely adversarial card — test-pressing forced at the confusion moment, delivery verified six of six — the model untangled her confusion five of six anyway. At a real confusion moment this model's default conduct IS the gold, and no card moves it; the entry is recorded as handled natively, its typed detection retained for routing and judge disclosure, its card explicitly not claimed as causal — the small-kernel law applied to the playbook's own table.

The wager promotion: the licence moves the learner; the beat stays unisolated. The last optional playbook question — whether the wager's fourth beat, staking the send on the learner's own check, buys anything in the learner — was run as a registered two-arm contrast at the first demand (licence on versus off, twelve cells pooled across the recorded and fresh runs, leaks zero across every licensed turn). On the registered intention-to-treat outcome the licence arm wins at the bar: learner check-engagement four of six versus one of six — licence-arm learners commit to the condition ("the email stays a draft unless we can show water got from that screen to the ceiling") or report the check performed ("Notebook's blank on a trace from the screen"), where harness-arm learners ask what would count. Send-deferral separates nothing (six of six in both arms — the three-beat harness already restrains the send everywhere; the licence's purchase is engagement, not restraint). The manipulation check keeps the claim honest: the full staked send appeared in only two of six licensed replies, and two of the licence arm's wins followed unstaked replies — so the promotion attaches to the standing licence, which sharpens the whole reply, and not cleanly to the fourth beat, whose isolated effect (per-protocol n=2, both the strongest ownership responses) stays an anecdote awaiting a forced-delivery build if it is ever wanted.

The last entry: flat boredom, proven twice over. The playbook's final unproven entry went through both halves of its admission in one registered pass. The crossed test at the ratified mid-dialogue bored moment (the endgame-drag plant was excluded rather than forced falsely — its gold has no matching card) came back right 5/5 versus wrong 0/5 on ruled conduct: every right-card reply re-engages the flat learner with a concrete oblique question ("what does the shed keep those tanks standing on, wood or stone?"), and every wrong-card reply performs the schedule's named error verbatim, one more careful recap of settled ground. And the detection hole that had kept the entry unpromotable — boredom in a learner too wordy to trigger the length collapse — closed as two closed-class shapes mined from the archive's eleven misses (the assent-opener sameness line; the clock-watching closer): replayed recall 66/66, zero pressure collisions, zero calm fires, and ten of ten on the fresh run's own bored plants as held-out shadow. The playbook census closes: six entries causally proven, one robust-native, the licence promoted with its price measured, and detection editions whose every claim carries corpus-wide plus held-out evidence. Repertoire growth for this generation is done; the next entry waits for a priced deficit, per the small-kernel rule (one-world bound on the bored crossed test stated; simulated learner throughout).

The section's one law, stated once. The due line (§6.23) gave the tutor a fact it could not see, and conduct improved. The move card gave it the moment, and repair delivery improved past the claim gate. Disclosure gave the judge the moment, and measurement improved. One currency — the learner's state and the schedule's time — spent on whichever role needs it, at the turn it applies; a standing prompt sets a rate where per-turn injection sets a timing, and every enforcement-shaped mechanism priced in this section subtracted. The model plays; the harness keeps time.

Status and caveats (per §5.12.6). Post-hoc exploratory throughout except the claim gate, which is pre-registered on the board card with its bar and null branch written before the runs; the v2 failure is reported beside the v3 pass, and the pass is taken at the comparison's weakest reading (the v3 tally is pre-adjudication; adjudication can only widen it). Scale limits: k=3 per arm at the original gate and n=1 per arm in the three-arm manner comparison; the Phase R paragraph lifts the world, persona, and seat-family limits to its stated pooled scope (k=5 sonnet, k=3 opus directional), while the simulated learner and the no-human-learning boundary stand throughout; the fold observation is bounded by two mid-run attritions. Phase R's per-world splits are tagger-sensitive at this k and the short-world coverage boundary (a pressure trigger under-arms on a fast world) is part of the claimed shape. The sol move-tagger carries a measured taste caveat (its tags re-litigated one gold ruling it had lost; human adjudication corrected six of 48 tags, moving the v2 tally 8→10 and 5→9 without changing the verdict). The disclosure result discloses *authored* states; live use would disclose the trigger's estimate and inherit its error. Repertoire-hole and quiet-state claims are about this tutor under prompting tested, not about the family. Provenance: worlds `config/drama-derivation/world-032-alder-row.yaml`, `world-033-alder-row-redoubt.yaml`; stress schedule (ratified, dual-provenance header) `config/drama-derivation/stress/world-033-stress-schedule.yaml`; switch and trigger `services/tutorStubMannerSwitch.js`, artifacts `config/manner-trigger/v1..v3`; guard catalog and shadow policy `services/tutorStubGuardDisposition.js` (v6), assessment `docs/tutor-stub-guard-catalog.md`; scorecards and reviews `scripts/score-manner-trigger.js`, `scripts/analyze-stance-contingency.js`, `scripts/summarize-tutor-stub-run.js`, `scripts/review-stress-bench.js`; runs and verdict files under `exports/tutor-stub-outcome/` (`misconception-gate-1..3`, `exp-stance-*`, `exp-shadow-*`, `stage5-*`, `stage5v3-switch`, `disclosure-judge*.json`, `stage5-tags.json`, `stage5v3-tags.json`); working synthesis `HOW-TO-BUILD-A-TUTOR.md`. No DB, rubric, abstract, or headline-N changes; no sections renumbered.

Neighbouring modules: Module 14 — Advice, Enforcement, and the Cost of Control;

Module 16 — Training One Teaching Move into a Small Model; Module 8 — The Ruler Is Part of the Research; Module 10 — The Model Still Sets the Ceiling.